

## 5.4 Roots, Multiplicity and Derivatives

Chapter 2 bounded the number of roots of a polynomial, and the first section of this chapter restated the bound as Corollary 5.1.4: a non-zero polynomial of degree  $n$  has at most  $n$  roots. The bound counts each root once, and so it misses something. The polynomial  $(x - 1)^2$  has degree 2 but only one root, and the root 1 is somehow “used twice”. This section makes “used twice” precise with the multiplicity of a root, sharpens the root bound, and then gives a test for repeated roots that needs no factoring: the formal derivative. The derivative behaves differently in characteristic  $p$ , and we say exactly where. The section ends with Taylor expansion of a polynomial, which turns out to be a dual basis from Chapter 4.

### 5.4.1 Multiplicity of a root

By the Factor Theorem (Lemma 2.7.1, restated in general as Theorem 5.1.3),  $c$  is a root of  $f$  exactly when  $x - c$  divides  $f$ . A repeated root should mean that a higher power of  $x - c$  divides  $f$ . So we measure how many factors  $x - c$  can be pulled out.

*The multiplicity of  $c$  counts how many times  $x - c$  divides  $f$ .*

#### Definition 5.4.1. Multiplicity of a Root

Let  $F$  be a field,  $f \in F[x]$  **non-zero** and  $c \in F$ . The **multiplicity of  $c$  as a root of  $f$** , written  $\text{mult}_c(f)$ , is the **largest**  $m \in \mathbb{N}$  such that  $(x - c)^m \mid f$ .

The root  $c$  is **simple** if  $\text{mult}_c(f) = 1$ , and **multiple** (or **repeated**) if  $\text{mult}_c(f) \geq 2$ .

In words: keep dividing  $f$  by  $x - c$  as long as the division leaves no remainder, and count the successful divisions. Divisibility is meant in the sense of Definition 5.1.1, so  $(x - c)^m \mid f$  says  $f = (x - c)^m g$  for some  $g \in F[x]$ .

There is something to check: a largest such  $m$  must exist. The value  $m = 0$  always works, since  $(x - c)^0 = 1$  divides every polynomial. And if  $f = (x - c)^m g$ , then  $g \neq 0$  because  $f \neq 0$ , so Theorem 0.8.3 gives  $\deg f = m + \deg g \geq m$ . The admissible  $m$  form a non-empty set of natural numbers bounded by  $\deg f$ , so it has a largest element. Moreover  $\text{mult}_c(f) \geq 1$  exactly when  $x - c \mid f$ , that is, exactly when  $c$  is a root. So multiplicity 0 means “not a root”.

The hypothesis  $f \neq 0$  is not decoration. Every power  $(x - c)^m$  divides the zero polynomial, since  $0 = (x - c)^m \cdot 0$ . There is no largest  $m$ , and the zero polynomial has no multiplicities.

A definition by “largest  $m$ ” is awkward to use directly, because to confirm  $m$  we must rule out all larger powers. The next lemma replaces it with a single test at  $c$ .

#### Lemma 5.4.1. Multiplicity via the Cofactor

Let  $f \in F[x]$  be non-zero,  $c \in F$  and  $m \in \mathbb{N}$ . Then  $\text{mult}_c(f) = m$  if and only if

$$f = (x - c)^m g \quad \text{for some } g \in F[x] \text{ with } g(c) \neq 0.$$

*Proof.*  $(\Rightarrow)$  Suppose  $\text{mult}_c(f) = m$ . Then  $f = (x - c)^m g$  for some  $g \in F[x]$ . If  $g(c) = 0$ , then  $g = (x - c)h$  by Lemma 2.7.1, so  $f = (x - c)^{m+1}h$ , contradicting the maximality of  $m$ . Hence  $g(c) \neq 0$ .

( $\Leftarrow$ ) Suppose  $f = (x - c)^m g$  with  $g(c) \neq 0$ . Then  $(x - c)^m \mid f$ , so  $\text{mult}_c(f) \geq m$ . Suppose, for a contradiction, that  $(x - c)^{m+1} \mid f$ , say  $f = (x - c)^{m+1} h$ . Then  $(x - c)^m g = (x - c)^m (x - c) h$ . Since  $(x - c)^m \neq 0$ , cancellation (Corollary 0.8.4) gives  $g = (x - c) h$ , and evaluating at  $c$  gives  $g(c) = 0$  by Theorem 0.8.5, a contradiction. Every higher power  $(x - c)^k$  with  $k > m + 1$  would also give  $(x - c)^{m+1} \mid f$ . Hence  $\text{mult}_c(f) = m$ . ■

The first payoff is that multiplicities add when polynomials multiply, just as exponents do.

### Theorem 5.4.2. Multiplicity of a Product

Let  $f, g \in F[x]$  be non-zero and  $c \in F$ . Then  $\text{mult}_c(fg) = \text{mult}_c(f) + \text{mult}_c(g)$ .

*Proof.* Let  $m = \text{mult}_c(f)$  and  $n = \text{mult}_c(g)$ . By Lemma 5.4.1,  $f = (x - c)^m u$  and  $g = (x - c)^n v$  with  $u(c) \neq 0$  and  $v(c) \neq 0$ . Then  $fg = (x - c)^{m+n} uv$ , and  $(uv)(c) = u(c)v(c) \neq 0$  by Theorem 0.8.5 and Theorem 0.7.1. By Lemma 5.4.1 again,  $\text{mult}_c(fg) = m + n$ . ■

### Example. Finding a Multiplicity by Repeated Division

Let  $f = x^4 - 2x^3 + 2x - 1 \in \mathbb{Q}[x]$ . Find the multiplicity of 1 and of  $-1$  as roots of  $f$ .

*Solution.* We have  $f(1) = 1 - 2 + 2 - 1 = 0$ , so we divide by  $x - 1$ . Long division gives

$$f = (x - 1)(x^3 - x^2 - x + 1).$$

The cofactor  $g_1 = x^3 - x^2 - x + 1$  has  $g_1(1) = 0$ , so we divide again:  $g_1 = (x - 1)(x^2 - 1)$ . The new cofactor  $g_2 = x^2 - 1$  has  $g_2(1) = 0$ , and  $g_2 = (x - 1)(x + 1)$ . Now  $g_3 = x + 1$  has  $g_3(1) = 2 \neq 0$ . Hence

$$f = (x - 1)^3(x + 1), \quad g_3(1) \neq 0,$$

and Lemma 5.4.1 gives  $\text{mult}_1(f) = 3$ . For  $-1$ , write  $f = (x + 1) \cdot (x - 1)^3$ , whose cofactor  $(x - 1)^3$  takes the value  $(-2)^3 = -8 \neq 0$  at  $-1$ . So  $\text{mult}_{-1}(f) = 1$ , a simple root. The multiplicities add up to  $3 + 1 = 4 = \deg f$ .

**Warning. Multiplicity is not visible from the set of roots.** The polynomials  $(x - 1)^3(x + 1)$  and  $(x - 1)(x + 1)^3$  have the same roots  $\{1, -1\}$ , the same degree and the same leading coefficient, but they are different polynomials. “The roots of  $f$ ” as a set loses information; the roots **with their multiplicities** do not, as the next theorem shows for polynomials that split.

*Quick check.* Over  $\mathbb{R}$ , what is the multiplicity of 0 as a root of  $x^3(x^2 + 1)$ ? Of 2 as a root of  $(x^2 - 4)^2$ ?

*Solution.* For  $x^3(x^2 + 1)$ , the cofactor  $x^2 + 1$  is  $1 \neq 0$  at 0, so the multiplicity is 3 by [Lemma 5.4.1](#). For  $(x^2 - 4)^2 = (x - 2)^2(x + 2)^2$ , the cofactor  $(x + 2)^2$  is  $16 \neq 0$  at 2, so the multiplicity is 2.

## 5.4.2 Counting roots with multiplicity

In [Example \(Finding a Multiplicity by Repeated Division\)](#) the multiplicities added up to exactly the degree. That cannot always happen:  $x^2 + 1 \in \mathbb{R}[x]$  has degree 2 and no real roots at all. The difference is whether  $f$  breaks into linear factors, so we give that a name.

### Definition 5.4.2. Splitting into Linear Factors

Let  $F$  be a field. A non-zero  $f \in F[x]$  of degree  $n$  **splits over**  $F$  if there are  $a \in F$  and  $a_1, \dots, a_n \in F$ , not necessarily distinct, with

$$f = a(x - a_1)(x - a_2) \cdots (x - a_n).$$

A non-zero constant splits (take  $n = 0$  and the empty product 1). The polynomial  $x^2 - 1 = (x - 1)(x + 1)$  splits over  $\mathbb{Q}$ . The polynomial  $x^2 - 2$  does not split over  $\mathbb{Q}$ , since a linear factor  $x - a_1$  with  $a_1 \in \mathbb{Q}$  would give a rational root, but it splits over  $\mathbb{R}$  as  $(x - \sqrt{2})(x + \sqrt{2})$ . The phrase **over**  $F$  carries weight: the  $a_i$  must lie in  $F$ .

It turns out that the multiplicities always fit inside the degree, and they fill it exactly when  $f$  splits:

### Theorem 5.4.3. Roots Counted with Multiplicity

Let  $F$  be a field and  $f \in F[x]$  non-zero. Let  $c_1, \dots, c_k$  be the distinct roots of  $f$  in  $F$ , with multiplicities  $m_1, \dots, m_k$ . Then

$$f = (x - c_1)^{m_1} \cdots (x - c_k)^{m_k} g$$

for some  $g \in F[x]$  with **no** root in  $F$ . Consequently

$$m_1 + \cdots + m_k \leq \deg f,$$

with equality if and only if  $f$  splits over  $F$ .

*Idea.* Peel off the roots one at a time. After removing  $(x - c_1)^{m_1}$ , the other roots  $c_j$  are still roots of the cofactor, because  $c_j - c_1 \neq 0$ , and by [Theorem 5.4.2](#) they keep their multiplicities. Whatever is left at the end has no roots, and its degree is exactly the gap between  $\deg f$  and  $\sum m_i$ . The gap is zero exactly when the leftover is a constant.

*Proof.* The set of roots is finite by Corollary 5.1.4, so the list  $c_1, \dots, c_k$  makes sense. We prove the following claim by induction on  $j \in \{0, 1, \dots, k\}$ :  $f = (x - c_1)^{m_1} \cdots (x - c_j)^{m_j} g_j$

for some non-zero  $g_j \in F[x]$  with

$$\text{mult}_{c_i}(g_j) = 0 \text{ for } i \leq j, \quad \text{mult}_{c_i}(g_j) = m_i \text{ for } i > j.$$

For  $j = 0$  take  $g_0 = f$ . Suppose the claim holds for some  $j < k$ . By Lemma 5.4.1 applied to  $g_j$ , we have  $g_j = (x - c_{j+1})^{m_{j+1}} g_{j+1}$  with  $g_{j+1}(c_{j+1}) \neq 0$ , which gives the product formula for  $j + 1$  and  $\text{mult}_{c_{j+1}}(g_{j+1}) = 0$ . For  $i \neq j + 1$ , the polynomial  $(x - c_{j+1})^{m_{j+1}}$  takes the value  $(c_i - c_{j+1})^{m_{j+1}} \neq 0$  at  $c_i$ , since the roots are distinct. So its multiplicity at  $c_i$  is 0 by Lemma 5.4.1 (with cofactor the polynomial itself), and Theorem 5.4.2 gives

$$\text{mult}_{c_i}(g_j) = 0 + \text{mult}_{c_i}(g_{j+1}).$$

This is 0 for  $i \leq j$  and  $m_i$  for  $i > j + 1$ , which completes the induction.

Put  $g = g_k$ . By the claim,  $\text{mult}_{c_i}(g) = 0$ , that is,  $g(c_i) \neq 0$ , for every  $i$ . If  $d \in F$  were a root of  $g$ , then  $f(d) = (d - c_1)^{m_1} \cdots (d - c_k)^{m_k} g(d) = 0$ , so  $d$  would be one of the  $c_i$ , which is impossible. Hence  $g$  has no root in  $F$ .

Since  $f \neq 0$ , also  $g \neq 0$ , and Theorem 0.8.3 gives  $\deg f = m_1 + \cdots + m_k + \deg g \geq m_1 + \cdots + m_k$ .

( $\Rightarrow$ ) If equality holds, then  $\deg g = 0$ , so  $g = a$  is a non-zero constant and  $f = a(x - c_1)^{m_1} \cdots (x - c_k)^{m_k}$  splits over  $F$ .

( $\Leftarrow$ ) Suppose  $f = a(x - a_1) \cdots (x - a_n)$  with  $n = \deg f$ . Group equal factors:  $f = a(x - d_1)^{n_1} \cdots (x - d_l)^{n_l}$  with  $d_1, \dots, d_l$  distinct and  $n_1 + \cdots + n_l = n$ . By the computation above, the cofactor  $a \prod_{s \neq r} (x - d_s)^{n_s}$  is non-zero at  $d_r$ , so  $\text{mult}_{d_r}(f) = n_r$  by Lemma 5.4.1. Every root  $d$  of  $f$  satisfies  $a \prod_s (d - d_s)^{n_s} = 0$ , so  $d$  is one of the  $d_r$ . Hence the roots of  $f$  are exactly  $d_1, \dots, d_l$ , and their multiplicities add up to  $n = \deg f$ . This proves the theorem. ■

The theorem sharpens Corollary 5.1.4, because the number  $k$  of distinct roots is at most  $m_1 + \cdots + m_k$ . It also shows that a split polynomial is determined by its leading coefficient and its roots with multiplicities. Whether a given polynomial splits depends on the field: the fifth-degree polynomial  $(x - 1)^3(x^2 + 1)$  has root multiplicities adding up to  $3 < 5$  over  $\mathbb{R}$ , but it splits over  $\mathbb{C}$  as  $(x - 1)^3(x - i)(x + i)$ . In the next section we prove that over  $\mathbb{C}$  **every** polynomial splits.

### 5.4.3 The formal derivative

How do we recognize a repeated root without factoring? Over  $\mathbb{R}$  calculus has an answer. At a simple root the graph of  $f$  crosses the axis with non-zero slope, while at a double root, as for  $(x - 1)^2$  at 1, it touches the axis and the slope is 0. We would like to use this over every field, including  $\mathbb{F}_p$ , where there are no limits and no slopes. So we keep the formula of the derivative and throw away the limit.

*The formal derivative is the power rule, applied coefficient by coefficient.*

#### Definition 5.4.3. Formal Derivative

Let  $F$  be a field and  $f = a_0 + a_1x + \cdots + a_nx^n \in F[x]$ . The **formal derivative** of  $f$  is

$$f' = a_1 + 2a_2x + 3a_3x^2 + \cdots + na_nx^{n-1} = \sum_{k=1}^n ka_kx^{k-1},$$

where  $ka_k$  is the integer multiple of  $a_k$  (the sum of  $k$  copies of  $a_k$ ). Higher derivatives are defined by  $f^{(0)} = f$  and  $f^{(k+1)} = (f^{(k)})'$ ; we also write  $f'' = f^{(2)}$ .

In words: each monomial  $a_kx^k$  becomes  $ka_kx^{k-1}$ , and constants disappear. This is the map  $D$  of Example (Differentiation), where we already saw that  $D: F[x] \rightarrow F[x]$  is linear. Over  $\mathbb{R}$  it agrees with the derivative of calculus on polynomial functions, since calculus gives the same power rule. By Lemma 0.7.4,  $ka_k = (k \cdot 1)a_k$ , so the integer  $k$  acts through the element  $k \cdot 1$  of  $F$ . **That element can be 0**, and this is the source of every difference between characteristic 0 and characteristic  $p$ .

Some examples. Over  $\mathbb{Q}$ ,  $(x^4 - 2x^3 + 2x - 1)' = 4x^3 - 6x^2 + 2$ . Over any field, the derivative of a constant is 0 and  $(x - c)' = 1$ . Over  $\mathbb{F}_3$ ,  $(x^3 + x)' = 3x^2 + 1 = 1$ , because  $3 \cdot 1 = 0$  there. Over  $\mathbb{F}_2$ ,  $(x^2)' = 2x = 0$ : a non-constant polynomial with derivative zero.

In characteristic 0 the last example cannot happen. If  $\deg f = n \geq 1$ , the coefficient of  $x^{n-1}$  in  $f'$  is  $(n \cdot 1)a_n$ , a product of two non-zero elements, so  $\deg f' = n - 1$  and in particular  $f' \neq 0$ . In characteristic  $p$  the same coefficient vanishes whenever  $p \mid n$ , because then  $n \cdot 1 = (n/p) \cdot (p \cdot 1) = 0$  by Lemma 0.7.4.

The derivative obeys the rules we know from calculus. The proofs cannot use limits, so they go through monomials.

#### Theorem 5.4.4. Rules for the Formal Derivative

Let  $F$  be a field,  $f, g \in F[x]$ ,  $a, b, c \in F$  and  $m \geq 1$  an integer. Then:

- (a)  $(af + bg)' = af' + bg'$ , and  $a' = 0$  for the constant polynomial  $a$ ;
- (b)  $(fg)' = f'g + fg'$  (product rule);
- (c)  $(g^m)' = mg^{m-1}g'$ ; in particular  $((x - c)^m)' = m(x - c)^{m-1}$ .

*Idea.* Both sides of the product rule are linear in  $f$  for fixed  $g$ , and linear in  $g$  for fixed  $f$ . So it is enough to check the rule when  $f$  and  $g$  are monomials, where it is the identity  $i + j = i + j$  for exponents. Checking on monomials, and extending by linearity, is the standard way to prove any identity about  $D$ .

*Proof.*

- (a) Linearity is Example (Differentiation). A constant has no coefficients in positions  $k \geq 1$ , so its derivative is 0.
- (b) First let  $f = x^i$  and  $g = x^j$  with  $i, j \in \mathbb{N}$ . If  $i, j \geq 1$ , then

$$(x^{i+j})' = (i + j)x^{i+j-1} = ix^{i-1}x^j + x^i jx^{j-1} = f'g + fg',$$

using  $(i + j) \cdot 1 = i \cdot 1 + j \cdot 1$  from Lemma 0.7.4. If  $i = 0$ , then  $f = 1$ ,  $f' = 0$  and both sides equal  $jx^{j-1}$  (or 0 when  $j = 0$ ); the case  $j = 0$  is symmetric.

Now let  $f = \sum_i a_i x^i$  and  $g = \sum_j b_j x^j$ . By Theorem 0.8.1,  $fg = \sum_{i,j} a_i b_j x^i x^j$ , a finite sum. Applying (a) and the monomial case,

$$\begin{aligned} (fg)' &= \sum_{i,j} a_i b_j ((x^i)' x^j + x^i (x^j)') \\ &= \left( \sum_i a_i (x^i)' \right) \left( \sum_j b_j x^j \right) + \left( \sum_i a_i x^i \right) \left( \sum_j b_j (x^j)' \right) \\ &= f'g + fg', \end{aligned}$$

where the middle equality regroups with the distributive laws, and the last uses (a) again.

(c) We use induction on  $m$ . For  $m = 1$  both sides are  $g'$ . If  $(g^m)' = mg^{m-1}g'$ , then by (b)

$$(g^{m+1})' = (g^m g)' = mg^{m-1}g'g + g^m g' = (m+1)g^m g'.$$

For  $g = x - c$  we have  $g' = 1$  by (a). This proves the theorem. ■

## 5.4.4 Repeated roots and the derivative

Now we can turn the calculus picture into a theorem valid over every field.

### Theorem 5.4.5. Repeated Root Test

Let  $F$  be a field,  $f \in F[x]$  non-zero and  $c \in F$ .

- (a)  $c$  is a multiple root of  $f$  if and only if  $f(c) = 0$  and  $f'(c) = 0$ .
- (b) Suppose  $\text{mult}_c(f) = m \geq 1$  and  $m \cdot 1 \neq 0$  in  $F$  (this holds automatically if  $F$  has characteristic 0). Then  $f' \neq 0$  and  $\text{mult}_c(f') = m - 1$ .

*Idea.* Write  $f = (x - c)g$  and differentiate:  $f' = g + (x - c)g'$ , so  $f'(c) = g(c)$ . The derivative at  $c$  reads off exactly the cofactor's value, which is what Lemma 5.4.1 needs. For (b), differentiate  $(x - c)^m g$  instead; the factor  $m$  appears, and it must not be 0 in  $F$ .

*Proof.*

- (a)  $(\Rightarrow)$  Suppose  $f = (x - c)^2 g$ . By Theorem 5.4.4 (b) and (c),  $f' = 2(x - c)g + (x - c)^2 g'$ , and evaluating at  $c$  gives  $f(c) = 0$  and  $f'(c) = 0$ .  
 $(\Leftarrow)$  Suppose  $f(c) = 0$  and  $f'(c) = 0$ . By Lemma 2.7.1,  $f = (x - c)g$  for some  $g \in F[x]$ . By Theorem 5.4.4,  $f' = g + (x - c)g'$ , so  $0 = f'(c) = g(c)$ . By Lemma 2.7.1 again,  $g = (x - c)h$ , so  $f = (x - c)^2 h$  and  $\text{mult}_c(f) \geq 2$ .
- (b) By Lemma 5.4.1,  $f = (x - c)^m g$  with  $g(c) \neq 0$ . By Theorem 5.4.4,

$$f' = m(x - c)^{m-1}g + (x - c)^m g' = (x - c)^{m-1} h, \quad h := mg + (x - c)g'.$$

Then  $h(c) = (m \cdot 1)g(c)$ , a product of two non-zero elements of  $F$ , so  $h(c) \neq 0$ . In particular  $h \neq 0$ , so  $f' \neq 0$  by Corollary 0.8.4, and  $\text{mult}_c(f') = m - 1$  by Lemma 5.4.1.

Part (a) holds over **every** field. Part (b) needs its hypothesis. Over  $\mathbb{F}_2$ , let  $f = x^2(x+1) = x^3+x^2$ . Then  $\text{mult}_0(f) = 2$ , and  $f' = 3x^2+2x = x^2$ , so  $\text{mult}_0(f') = 2$ , not 1. Here  $m \cdot 1 = 2 \cdot 1 = 0$ , and the factor  $m$  killed the term that should have lowered the multiplicity.

Part (a) finds a repeated root at a known point  $c$ . To decide whether **any** repeated root exists, without knowing the roots, we use the gcd of  $f$  and  $f'$ , which the Euclidean algorithm (Theorem 5.2.6) computes by division alone. Repeated roots are a special case of repeated irreducible factors:  $c$  is a multiple root exactly when the irreducible polynomial  $x - c$  appears squared.

#### Corollary 5.4.6. Repeated Factors and $\gcd(f, f')$

Let  $F$  be a field and  $f \in F[x]$  non-constant.

- (a) (**Any field.**) If  $p \in F[x]$  is irreducible and  $p^2 \mid f$ , then  $p \mid \gcd(f, f')$ . Hence, if  $\gcd(f, f') = 1$ , then no irreducible polynomial divides  $f$  twice; in particular  $f$  has no multiple root in  $F$ .
- (b) (**Characteristic 0.**) If  $F$  has characteristic 0 and  $\gcd(f, f') \neq 1$ , then  $p^2 \mid f$  for some irreducible  $p \in F[x]$ .

*Idea.* For (a), differentiate  $p^2h$ : every term still contains  $p$ . For (b), run it backwards. An irreducible  $p$  dividing both  $f = pg$  and  $f' = p'g + pg'$  must divide  $p'g$ . If  $p$  does not divide  $p'$ , then Euclid's lemma pushes  $p$  into  $g$ , and  $p^2 \mid f$ . The only question is whether  $p \mid p'$  can happen, and that is exactly where the characteristic enters.

*Proof.*

- (a) Write  $f = p^2h$ . By Theorem 5.4.4,  $f' = 2pp'h + p^2h' = p(2p'h + ph')$ . So  $p$  is a common divisor of  $f$  and  $f'$ , and  $p \mid \gcd(f, f')$  by Theorem 5.2.3. If  $\gcd(f, f') = 1$ , this would give  $p \mid 1$ , which is impossible because  $p$  is non-constant (Definition 5.3.1) and a non-zero multiple of  $p$  has degree at least  $\deg p \geq 1$  by Theorem 0.8.3. If  $c$  were a multiple root, then  $(x - c)^2 \mid f$ . But  $x - c$  is irreducible (Definition 5.3.1), because a product of two non-constant polynomials has degree at least 2. This is again impossible.
- (b) Let  $d = \gcd(f, f') \neq 1$ . Since  $d \mid f$  and  $f \neq 0$ ,  $d \neq 0$ ; as  $d$  is monic and  $d \neq 1$ ,  $\deg d \geq 1$ . By Theorem 5.3.5,  $d$  has a monic irreducible factor  $p$ . Then  $p \mid f$  and  $p \mid f'$  (Theorem 5.2.3). Write  $f = pg$ . By Theorem 5.4.4,  $f' = p'g + pg'$ , so  $p$  divides  $f' - pg' = p'g$ .

Since  $F$  has characteristic 0 and  $\deg p \geq 1$ , we showed after Definition 5.4.3 that  $p' \neq 0$  and  $\deg p' = \deg p - 1$ . A non-zero multiple of  $p$  has degree at least  $\deg p$ , so  $p \nmid p'$ . By Euclid's lemma (Theorem 5.3.3),  $p \mid g$ . Hence  $p^2 \mid pg = f$ . This proves the corollary.

**Warning.** In characteristic  $p$ , a non-constant polynomial can have derivative 0. Over  $\mathbb{F}_p$ ,  $(x^p)' = (p \cdot 1)x^{p-1} = 0$ . The proof of (b) used that an irreducible factor  $q$  of  $f$  has  $q' \neq 0$  of smaller degree, so that  $q \nmid q'$ ; in characteristic  $p$  that step is no longer automatic, because  $q'$  may be 0. So the correct statements are:  $\gcd(f, f') = 1$  implies no repeated irreducible factor, over **any** field; the converse is proved here **only in characteristic 0**. Do not use “repeated factor if and only if  $\gcd(f, f') \neq 1$ ” over a field of characteristic  $p$  without checking that no irreducible factor of  $f$  has derivative 0.

**Remark.** The converse really can fail in characteristic  $p$ , though not over the fields  $\mathbb{F}_p$ . Over the field of rational functions in a variable  $t$  with coefficients in  $\mathbb{F}_p$ , which this book does not construct, the polynomial  $x^p - t$  is irreducible, so it has no repeated factor, yet its derivative is 0.

### Example. Detecting a Repeated Root with a gcd

Decide whether  $f = x^4 - 2x^3 + 2x - 1 \in \mathbb{Q}[x]$  has a repeated irreducible factor, using only  $\gcd(f, f')$ . Does  $f$  have a multiple root in  $\mathbb{C}$ ?

**Solution.** Here  $f' = 4x^3 - 6x^2 + 2$ . Run the Euclidean algorithm (Theorem 5.2.6). Dividing  $f$  by  $f'$ ,

$$f = \left(\frac{1}{4}x - \frac{1}{8}\right)f' + \left(-\frac{3}{4}x^2 + \frac{3}{2}x - \frac{3}{4}\right),$$

and the remainder is  $-\frac{3}{4}(x^2 - 2x + 1)$ . Dividing  $f'$  by  $x^2 - 2x + 1$ ,

$$f' = (4x + 2)(x^2 - 2x + 1) + 0.$$

(We divided by the monic multiple of the remainder; scaling a divisor by a non-zero constant changes only the quotient, so the next remainder is still 0.) The last non-zero remainder, made monic, is the gcd:  $\gcd(f, f') = x^2 - 2x + 1 = (x - 1)^2 \neq 1$ . By [Corollary 5.4.6](#) (b), since  $\mathbb{Q}$  has characteristic 0,  $f$  has a repeated irreducible factor. Indeed  $x - 1$  divides the gcd, so 1 is a common root of  $f$  and  $f'$ , and by [Theorem 5.4.5](#) (a) it is a multiple root. This matches [Example \(Finding a Multiplicity by Repeated Division\)](#), where  $\text{mult}_1(f) = 3$ , and [Theorem 5.4.5](#) (b) gives  $\text{mult}_1(f') = 2$ , consistent with the factor  $(x - 1)^2$  of the gcd.

So yes,  $f$  has a multiple root in  $\mathbb{C}$ , namely  $1 \in \mathbb{Q}$ . The gcd test is useful over  $\mathbb{C}$  even for polynomials with rational coefficients: the Euclidean algorithm uses only divisions in  $\mathbb{Q}[x]$ , and division with remainder is unique (Theorem 5.1.1), so running it in  $\mathbb{C}[x]$  produces the same remainders and the same gcd. Had the gcd been 1, part (a) over  $\mathbb{C}$  would have ruled out every multiple complex root, without our finding a single root.

**Quick check.** Over  $\mathbb{R}$ ,  $f = x^3$  satisfies  $f(0) = f'(0) = 0$ . Does [Theorem 5.4.5](#) say that 0 has multiplicity 2?



*Solution.* No. Part (a) says only that the multiplicity is **at least** 2. The actual multiplicity is 3. Part (b) applies, since  $\mathbb{R}$  has characteristic 0:  $\text{mult}_0(f') = \text{mult}_0(3x^2) = 2 = 3 - 1$ . To pin the multiplicity down with derivatives, find the first  $k$  with  $f^{(k)}(0) \neq 0$ : here  $f''(0) = 0$  and  $f'''(0) = 6 \neq 0$ , so  $k = 3$ .

### 5.4.5 Taylor expansion of a polynomial

The answer to the check suggests that, in characteristic 0, the derivatives of  $f$  at  $c$  know everything about  $f$  near  $c$ . Calculus makes this precise with Taylor's formula. For polynomials the formula is exact, and it has a clean linear-algebra meaning: the numbers  $f^{(k)}(c)/k!$  are the coordinates of  $f$  in the basis of powers of  $x - c$ , so the maps  $f \mapsto f^{(k)}(c)/k!$  form a dual basis in the sense of Definition 4.1.3. In Example (Dual Bases in  $\mathbb{R}[x]_{\leq 2}$ ) we saw this for  $c = 0$  and  $n = 2$ .

We need one derivative computation. For integers  $0 \leq j \leq k$ , repeated use of Theorem 5.4.4 (c) and (a) gives

$$((x - c)^k)^{(j)} = k(k - 1) \cdots (k - j + 1) (x - c)^{k-j},$$

by induction on  $j$ , and  $((x - c)^k)^{(j)} = 0$  for  $j > k$ , since the  $k$ -th derivative is the constant  $k!$ . Evaluating at  $c$ , every term with  $j < k$  vanishes because of the factor  $(x - c)^{k-j}$ . Hence

$$((x - c)^k)^{(j)}(c) = \begin{cases} k! \cdot 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

#### Theorem 5.4.7. Taylor Expansion of a Polynomial

Let  $F$  be a field,  $n \in \mathbb{N}$  and  $c \in F$ .

- (a) The list  $\mathcal{B}_c = (1, x - c, (x - c)^2, \dots, (x - c)^n)$  is a basis of  $F[x]_{\leq n}$ .
- (b) Suppose  $F$  has characteristic 0, or characteristic  $p > n$ . For  $k = 0, \dots, n$  define  $\varphi_k: F[x]_{\leq n} \rightarrow F$  by  $\varphi_k(f) = \frac{f^{(k)}(c)}{k!}$ . Then  $(\varphi_0, \dots, \varphi_n)$  is the dual basis of  $\mathcal{B}_c$ , and every  $f \in F[x]_{\leq n}$  satisfies

$$f = \sum_{k=0}^n \frac{f^{(k)}(c)}{k!} (x - c)^k.$$

*Idea.* ① The powers of  $x - c$  have degrees  $0, 1, \dots, n$ , so no combination of them can cancel down to zero; there are  $n + 1$  of them, so counting finishes (a). ② The computation above says exactly  $\varphi_j((x - c)^k) = \delta_{jk}$ , once we may divide by  $k!$ . ③ A list of functionals with the  $\delta$  property is the dual basis, and the dual basis expansion  $\mathbf{v} = \sum \varphi_k(\mathbf{v}) \mathbf{v}_k$  of Theorem 4.1.3 is Taylor's formula.

*Proof.*

- (a) Let  $b_0 + b_1(x - c) + \cdots + b_n(x - c)^n = 0$ , and suppose not all  $b_k$  are zero. Let  $k$  be the largest index with  $b_k \neq 0$ . By Theorem 0.8.3,  $b_k(x - c)^k$  has degree  $k$ , and each earlier term has degree less than  $k$ , so by Theorem 0.8.2 the sum has degree  $k \geq 0$ .

This contradicts the sum being 0, of degree  $-\infty$ . Hence  $\mathcal{B}_c$  is linearly independent. It has  $n + 1$  elements and  $\dim F[x]_{\leq n} = n + 1$  by Example (Standard Bases), so  $\mathcal{B}_c$  is a basis by Theorem 1.6.4 (a).

- (b) First,  $k! \cdot 1 \neq 0$  in  $F$  for  $0 \leq k \leq n$ . By Lemma 0.7.4,  $k! \cdot 1 = (1 \cdot 1)(2 \cdot 1) \cdots (k \cdot 1)$ . In characteristic 0 each factor is non-zero by Definition 0.7.2; in characteristic  $p > n$ , each factor  $j \cdot 1$  with  $1 \leq j \leq k < p$  is non-zero by the minimality of  $p$ . A product of non-zero elements is non-zero (Theorem 0.7.1). So  $\varphi_k$  is defined.

Each  $\varphi_k$  is linear:  $f \mapsto f^{(k)}$  is linear as a composite of copies of  $D$  (Example (Differentiation)), evaluation at  $c$  is linear by Theorem 0.8.5, and so is multiplication by  $(k!)^{-1}$ . The derivatives of a polynomial of degree at most  $n$  have degree at most  $n$ , so  $\varphi_k \in F[x]_{\leq n}^*$ . By the computation before the theorem,

$$\varphi_j((x - c)^k) = \frac{1}{j!}((x - c)^k)^{(j)}(c) = \delta_{jk} \quad (0 \leq j, k \leq n).$$

By the uniqueness in Theorem 4.1.3 (a),  $(\varphi_0, \dots, \varphi_n)$  is the dual basis of  $\mathcal{B}_c$ . The first formula of Theorem 4.1.3 (c),  $f = \sum_k \varphi_k(f)(x - c)^k$ , is the Taylor expansion. This proves the theorem. ■

### Example. Re-centering a Cubic

Expand  $f = x^3 - 2x + 5 \in \mathbb{Q}[x]$  in powers of  $x - 1$ .

*Solution.* The derivatives are  $f' = 3x^2 - 2$ ,  $f'' = 6x$ ,  $f''' = 6$ . At  $c = 1$ :  $f(1) = 4$ ,  $f'(1) = 1$ ,  $f''(1) = 6$ ,  $f'''(1) = 6$ . By Theorem 5.4.7 (b), since  $\mathbb{Q}$  has characteristic 0,

$$\begin{aligned} f &= 4 + 1 \cdot (x - 1) + \frac{6}{2}(x - 1)^2 + \frac{6}{6}(x - 1)^3 \\ &= 4 + (x - 1) + 3(x - 1)^2 + (x - 1)^3. \end{aligned}$$

Check by expanding:  $(x - 1)^3 + 3(x - 1)^2 + (x - 1) + 4$  equals

$$(x^3 - 3x^2 + 3x - 1) + (3x^2 - 6x + 3) + (x - 1) + 4 = x^3 - 2x + 5.$$

The expansion also reads off root information at 1: the constant term  $f(1) = 4 \neq 0$ , so 1 is not a root. In general,  $\text{mult}_c(f)$  is the index of the first non-zero coordinate of  $f$  in the basis  $\mathcal{B}_c$  (over any field, by part (a); in characteristic 0 these coordinates are the Taylor coefficients), because  $f = (x - c)^m(b_m + b_{m+1}(x - c) + \cdots)$  and the bracket is  $b_m$  at  $c$ .

**Warning. Taylor's formula needs  $k! \neq 0$ .** Over  $\mathbb{F}_2$ , take  $f = x^2$  and  $c = 0$ . Then  $f' = 2x = 0$ , so **every** derivative of  $f$ , at every point, is 0, even though  $f \neq 0$ . No formula in the values  $f^{(k)}(0)$  can recover the coefficient 1 of  $x^2$ . Part (a) of Theorem 5.4.7 still holds over  $\mathbb{F}_2$ :  $x^2$  has coordinates  $(0, 0, 1)$  in  $(1, x, x^2)$ . What fails is that these coordinates are derivatives divided by  $k!$ .

The two halves of this section meet in Chapter 8, where a matrix is diagonalizable exactly when its minimal polynomial splits and has no repeated roots; the gcd test above is how such a condition is checked in practice. Before that, the next section proves that over  $\mathbb{C}$  the sum of the multiplicities always equals the degree.

## 5.4.6 Exercises

### 5.4.6.1 A. Check your understanding

#### Exercise. A1

- (a) Define the multiplicity of  $c$  as a root of a non-zero  $f \in F[x]$ . Why is  $f \neq 0$  required?
- (b) True or false: if  $f(c) = f'(c) = 0$ , then  $c$  has multiplicity exactly 2. Justify your answer.
- (c) True or false: over every field,  $f' = 0$  implies that  $f$  is constant. Justify your answer.
- (d) True or false: if  $f \in \mathbb{R}[x]$  has degree 4 and the multiplicities of its real roots add up to 4, then  $f$  splits over  $\mathbb{R}$ . Justify your answer.
- (e) True or false: if  $f \in \mathbb{Q}[x]$  is non-constant and  $\gcd(f, f') \neq 1$ , then  $f$  has a repeated irreducible factor. Justify your answer.
- (f) Name the dual basis (Theorem 4.1.3) of  $(1, x - c, \dots, (x - c)^n)$  in  $\mathbb{R}[x]_{\leq n}$ .

*Solution.*

- (a) It is the largest  $m \in \mathbb{N}$  with  $(x - c)^m \mid f$  (Definition 5.4.1). For  $f = 0$ , every power divides  $f$ , so no largest  $m$  exists.
- (b) False. For  $f = x^3$  and  $c = 0$ ,  $f(0) = f'(0) = 0$ , but  $\text{mult}_0(f) = 3$ . Theorem 5.4.5 (a) only gives multiplicity at least 2.
- (c) False. Over  $\mathbb{F}_2$ ,  $(x^2)' = 2x = 0$ . (In characteristic 0 it is true:  $\deg f' = \deg f - 1$  whenever  $\deg f \geq 1$ .)
- (d) True. By Theorem 5.4.3, the sum of multiplicities equals the degree if and only if  $f$  splits over  $\mathbb{R}$ .
- (e) True.  $\mathbb{Q}$  has characteristic 0, so this is Corollary 5.4.6 (b).
- (f)  $\varphi_k(f) = f^{(k)}(c)/k!$  for  $k = 0, \dots, n$ , by Theorem 5.4.7 (b);  $\mathbb{R}$  has characteristic 0.

### 5.4.6.2 B. Practice

#### Exercise. B1: Multiplicities over different fields

- (a) Let  $f = x^5 - 3x^4 + 4x^3 - 4x^2 + 3x - 1$ . Find  $\text{mult}_1(f)$  in  $\mathbb{Q}[x]$ . Hence decide whether  $f$  splits over  $\mathbb{R}$ , and whether it splits over  $\mathbb{C}$ .
- (b) In  $\mathbb{F}_2[x]$ , find the roots of  $g = x^4 + x^2$  and their multiplicities. Does  $g$  split over  $\mathbb{F}_2$ ?

*Solution.*

- (a) Here  $f(1) = 1 - 3 + 4 - 4 + 3 - 1 = 0$ . Dividing by  $x - 1$  repeatedly,

$$\begin{aligned} f &= (x - 1)(x^4 - 2x^3 + 2x^2 - 2x + 1), \\ x^4 - 2x^3 + 2x^2 - 2x + 1 &= (x - 1)(x^3 - x^2 + x - 1), \end{aligned}$$

and  $x^3 - x^2 + x - 1 = (x - 1)(x^2 + 1)$ . Since  $x^2 + 1$  takes the value  $2 \neq 0$  at 1, [Lemma 5.4.1](#) gives  $f = (x - 1)^3(x^2 + 1)$  and  $\text{mult}_1(f) = 3$ . Over  $\mathbb{R}$ ,  $x^2 + 1$  has no root ( $x^2 + 1 \geq 1$ ), so the only real root of  $f$  is 1, and  $3 < 5 = \deg f$ . By [Theorem 5.4.3](#),  $f$  does not split over  $\mathbb{R}$ . Over  $\mathbb{C}$ ,  $x^2 + 1 = (x - i)(x + i)$ , so  $f = (x - 1)^3(x - i)(x + i)$  splits.

- (b) In  $\mathbb{F}_2[x]$ ,  $x^4 + x^2 = x^2(x^2 + 1)$ , and  $x^2 + 1 = (x + 1)^2$  because  $(x + 1)^2 = x^2 + 2x + 1 = x^2 + 1$ . Hence  $g = x^2(x + 1)^2 = (x - 0)^2(x - 1)^2$ , using  $-1 = 1$ . The cofactor  $(x + 1)^2$  is 1 at 0, and  $x^2$  is 1 at 1, so  $\text{mult}_0(g) = \text{mult}_1(g) = 2$  by [Lemma 5.4.1](#). These are the only elements of  $\mathbb{F}_2$ , the multiplicities add to  $4 = \deg g$ , and  $g$  splits over  $\mathbb{F}_2$ .

### Exercise. B2: A gcd test

Decide whether  $f = x^4 - 2x^2 + 1 \in \mathbb{Q}[x]$  has a repeated root, by computing  $\gcd(f, f')$  with the Euclidean algorithm. Hence factor  $f$  and give all multiplicities.

*Solution.* We have  $f' = 4x^3 - 4x$ . Dividing,  $f = \frac{1}{4}x \cdot f' + (-x^2 + 1)$ , since  $\frac{1}{4}x(4x^3 - 4x) = x^4 - x^2$ . Next,  $f' = (-4x)(-x^2 + 1) + 0$ . So the last non-zero remainder is  $-x^2 + 1$ , and by [Theorem 5.2.6](#)  $\gcd(f, f') = x^2 - 1 \neq 1$ . Its roots 1 and  $-1$  are common roots of  $f$  and  $f'$ , since  $x^2 - 1 = (x - 1)(x + 1)$  divides both. By [Theorem 5.4.5](#) (a), 1 and  $-1$  are multiple roots of  $f$ .

To factor, note that  $(x^2 - 1)^2 = x^4 - 2x^2 + 1 = f$ , so  $f = (x - 1)^2(x + 1)^2$ . By [Lemma 5.4.1](#),  $\text{mult}_1(f) = 2$  (cofactor  $(x + 1)^2$ , value 4 at 1) and  $\text{mult}_{-1}(f) = 2$ .

### Exercise. B3: The multiplicity of 1

Let  $n \geq 2$  and  $f = x^n - nx + n - 1 \in \mathbb{Q}[x]$ . Prove that 1 is a root of  $f$  of multiplicity exactly 2.

*Solution.* We have  $f(1) = 1 - n + n - 1 = 0$ ,  $f' = nx^{n-1} - n$ , so  $f'(1) = 0$ , and  $f'' = n(n - 1)x^{n-2}$ , so  $f''(1) = n(n - 1) \neq 0$  since  $n \geq 2$ . In particular  $f \neq 0$ . By [Theorem 5.4.5](#) (a),  $m := \text{mult}_1(f) \geq 2$ . Since  $\mathbb{Q}$  has characteristic 0, part (b) gives  $\text{mult}_1(f') = m - 1 \geq 1$ . Apply (a) to  $f' \neq 0$ : since  $f''(1) \neq 0$ , 1 is not a multiple root of  $f'$ , so  $m - 1 = 1$ . Hence  $m = 2$ .

### 5.4.6.3 C. Going deeper

#### Exercise. C1: Real roots of the derivative

Let  $f \in \mathbb{R}[x]$  have degree  $n \geq 2$  and split over  $\mathbb{R}$ . Prove that  $f'$  splits over  $\mathbb{R}$ . You may use **Rolle's theorem** from calculus: if  $a < b$  are real and  $f(a) = f(b) = 0$ , then  $f'(\xi) = 0$  for

some  $\xi$  with  $a < \xi < b$ .

*Hint: count the roots of  $f'$  with multiplicity, and compare with  $\deg f'$ .*

*Solution.* Let  $c_1 < c_2 < \cdots < c_k$  be the distinct real roots of  $f$ , with multiplicities  $m_1, \dots, m_k$ . Since  $f$  splits over  $\mathbb{R}$ , [Theorem 5.4.3](#) gives  $m_1 + \cdots + m_k = n$ .

By Rolle's theorem, for each  $i = 1, \dots, k-1$  there is  $\xi_i$  with  $c_i < \xi_i < c_{i+1}$  and  $f'(\xi_i) = 0$ . These  $k-1$  numbers are distinct, since they lie in disjoint open intervals, and none of them is a  $c_j$ . Moreover, each  $c_i$  with  $m_i \geq 2$  is a root of  $f'$  of multiplicity  $m_i - 1$ , by [Theorem 5.4.5](#) (b) (characteristic 0); for  $m_i = 1$  the count  $m_i - 1 = 0$  is also correct.

Since  $\deg f = n \geq 1$  and  $\mathbb{R}$  has characteristic 0,  $\deg f' = n - 1$ , so  $f' \neq 0$ . Adding the multiplicities of the distinct roots  $\xi_1, \dots, \xi_{k-1}, c_1, \dots, c_k$  of  $f'$  gives at least

$$(k-1) + \sum_{i=1}^k (m_i - 1) = (k-1) + n - k = n - 1 = \deg f'.$$

By [Theorem 5.4.3](#) the sum of the multiplicities of all roots of  $f'$  is at most  $\deg f'$ , so it equals  $\deg f'$ , and  $f'$  splits over  $\mathbb{R}$ .

**Exercise. C2:**  $x^p - x$  over  $\mathbb{F}_p$

Let  $p$  be a prime and  $f = x^p - x \in \mathbb{F}_p[x]$ .

- Compute  $f'$  and deduce that  $f$  has no repeated irreducible factor.
- Using Exercise (C2: A non-zero polynomial vanishing everywhere), deduce that  $x^p - x = \prod_{a \in \mathbb{F}_p} (x - a)$ .
- Explain why the argument of (a) gives nothing for  $g = x^p - 1$ . Then prove that  $g = (x - 1)^p$  in  $\mathbb{F}_p[x]$ .

*Hint for (c): for  $0 < k < p$ , use the integer identity  $k \binom{p}{k} = p \binom{p-1}{k-1}$ .*

*Solution.*

- By [Definition 5.4.3](#),  $f' = px^{p-1} - 1 = -1$ , since  $p \cdot 1 = 0$  in  $\mathbb{F}_p$ . Every common divisor of  $f$  and  $-1$  divides 1, so  $\gcd(f, f') = 1$ , and [Corollary 5.4.6](#) (a), which holds over every field, shows  $f$  has no repeated irreducible factor.
- By Exercise (C2: A non-zero polynomial vanishing everywhere) (c), every  $a \in \mathbb{F}_p$  is a root of  $f$ , so each of the  $p$  elements has multiplicity at least 1. The multiplicities add up to at least  $p = \deg f$ , and at most  $\deg f$  by [Theorem 5.4.3](#). So they add up to exactly  $p$ , and each is 1. By [Theorem 5.4.3](#),  $f = \prod_{b \in \mathbb{F}_p} (x - b) g$  with  $\deg g = p - p = 0$ , so  $g$  is a constant; comparing leading coefficients ([Theorem 0.8.3](#)),  $g = 1$ .
- Here  $g' = px^{p-1} = 0$ , so  $\gcd(g, g') = g \neq 1$  (the monic generator of the ideal generated by  $g$  and 0). Part (a) of [Corollary 5.4.6](#) needs  $\gcd = 1$ , so it says nothing, and part (b) is only available in characteristic 0. Still,  $g(1) = 0$  and  $g'(1) = 0$ , so 1 is a multiple root by [Theorem 5.4.5](#) (a).

For the factorization, expand with the binomial theorem (valid in the commutative ring  $\mathbb{F}_p[x]$ , [Theorem 0.8.1](#)):  $(x - 1)^p = \sum_{k=0}^p \binom{p}{k} (-1)^{p-k} x^k$ , with the integers  $\binom{p}{k}$  acting

as integer multiples. For  $0 < k < p$ , the identity  $k \binom{p}{k} = p \binom{p-1}{k-1}$  and Lemma 0.7.4 give  $(k \cdot 1) \left( \binom{p}{k} \cdot 1 \right) = (p \cdot 1) \left( \binom{p-1}{k-1} \cdot 1 \right) = 0$  in  $\mathbb{F}_p$ . Since  $0 < k < p$ ,  $k \cdot 1 \neq 0$  by Definition 0.7.2, so  $\binom{p}{k} \cdot 1 = 0$  (Theorem 0.7.1). Hence  $(x-1)^p = x^p + (-1)^p$ . For odd  $p$  this is  $x^p - 1$ ; for  $p = 2$  it is  $x^2 + 1 = x^2 - 1$ . So  $g = (x-1)^p$ , and  $\text{mult}_1(g) = p$  by Lemma 5.4.1.

### Exercise. C3: Removing repeated factors

Let  $F$  be a field and  $f = a p_1^{e_1} \cdots p_r^{e_r} \in F[x]$ , where  $a \in F \setminus \{0\}$ , the  $p_i$  are **distinct** monic irreducible polynomials and  $e_i \geq 1$ .

- (a) Suppose  $F$  has characteristic 0. Prove that  $\gcd(f, f') = p_1^{e_1-1} \cdots p_r^{e_r-1}$ , and deduce that  $f / \gcd(f, f') = a p_1 \cdots p_r$ .
- (b) Over  $\mathbb{F}_2$ , let  $f = x^2(x+1)$ . Compute  $f / \gcd(f, f')$  and show that the conclusion of (a) fails.

*Hint for (a): write  $D = p_1^{e_1-1} \cdots p_r^{e_r-1}$ . Show  $D \mid \gcd(f, f')$ ; for the reverse, show  $p_i^{e_i} \nmid f'$  using Euclid's lemma.*

*Solution.*

- (a) Let  $D = \prod_i p_i^{e_i-1}$  and  $d = \gcd(f, f')$ . By Theorem 5.4.4 (b) and (c), applied repeatedly,

$$f' = a \sum_{i=1}^r e_i p_i^{e_i-1} p'_i \prod_{j \neq i} p_j^{e_j}.$$

Every term is divisible by  $D$ , and so is  $f$ . Hence  $D \mid d$  by Theorem 5.2.3.

Conversely,  $d \mid f$ , so  $f = dh$ . Factoring  $d$  and  $h$  into monic irreducibles and comparing with the factorization of  $f$  by the uniqueness in Theorem 5.3.5,  $d = \prod_i p_i^{k_i}$  with  $0 \leq k_i \leq e_i$  (as  $d$  is monic). Suppose  $k_i = e_i$  for some  $i$ . Then  $p_i^{e_i} \mid f'$ . In the formula for  $f'$  every term with index  $j \neq i$  contains  $p_i^{e_i}$ , so  $p_i^{e_i}$  divides the remaining term  $a e_i p_i^{e_i-1} p'_i \prod_{j \neq i} p_j^{e_j}$ . Canceling  $p_i^{e_i-1}$  (Corollary 0.8.4),  $p_i \mid (a e_i) p'_i \prod_{j \neq i} p_j^{e_j}$ . Here  $a e_i = a(e_i \cdot 1)$  is a non-zero constant (characteristic 0),  $p_i \nmid p'_i$  because  $p'_i \neq 0$  has smaller degree, and  $p_i \nmid p_j$  for  $j \neq i$  because both are monic irreducible and distinct. Applying Theorem 5.3.3 repeatedly gives a contradiction. Hence  $k_i \leq e_i - 1$  for all  $i$ , so  $d \mid D$ . Both are monic and divide each other, so they have the same degree and  $d = D$ . Finally  $f = D \cdot a p_1 \cdots p_r$ , so  $f/d = a p_1 \cdots p_r$ .

- (b) In  $\mathbb{F}_2[x]$ ,  $f = x^3 + x^2$  and  $f' = 3x^2 + 2x = x^2$ . Since  $x^2 \mid f$ ,  $\gcd(f, f') = x^2$ , and  $f / \gcd(f, f') = x + 1$ . The formula of (a) would predict  $\gcd(f, f') = x$  and  $f / \gcd(f, f') = x(x+1)$ . The irreducible factor  $x$  has been lost. The step that fails is “ $a e_i \neq 0$ ”: here  $e_1 = 2$  and  $2 \cdot 1 = 0$  in  $\mathbb{F}_2$ .